

OpenStax Calculus Volume 3

Section 5.6 Exercises: Calculating Centers of Mass and Moments of Inertia

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In the following exercises, the region R occupied by a lamina is shown in a graph. Find the mass of R with the density function ρ .

1. R is the triangular region with vertices $(0, 0)$, $(0, 3)$, and $(6, 0)$; $\rho(x, y) = xy$.
[Figure: A right triangle bounded by the x and y axes and the line $y = \text{negative } x/2 + 3$.]
2. R is the triangular region with vertices $(0, 0)$, $(1, 1)$, $(0, 5)$; $\rho(x, y) = x + y$.
[Figure: A triangle bounded by the y axis, the line $x = y$, and the line $y = \text{negative } 4x + 5$.]
3. R is the rectangular region with vertices $(0, 0)$, $(0, 3)$, $(6, 3)$, and $(6, 0)$; $\rho(x, y) = \sqrt{xy}$.
[Figure: A rectangle bounded by the x and y axes and the lines $x = 6$ and $y = 3$.]
4. R is the rectangular region with vertices $(0, 1)$, $(0, 3)$, $(3, 3)$, and $(3, 1)$; $\rho(x, y) = x^2y$.
[Figure: A rectangle bounded by the y axis, the lines $y = 1$ and 3 , and the line $x = 3$.]
5. R is the trapezoidal region determined by the lines $y = -\frac{1}{4}x + \frac{5}{2}$, $y = 0$, $y = 2$, and $x = 0$; $\rho(x, y) = 3xy$.
[Figure: A trapezoid bounded by the x and y axes, the line $y = 2$, and the line $y = \text{negative } x/4 + 2.5$.]
6. R is the trapezoidal region determined by the lines $y = 0$, $y = 1$, $y = x$, and $y = -x + 3$; $\rho(x, y) = 2x + y$.
[Figure: A trapezoid bounded by the x axis, the line $y = 1$, the line $y = x$, and the line $y = \text{negative } x + 3$.]
7. R is the disk of radius 2 centered at $(1, 2)$; $\rho(x, y) = x^2 + y^2 - 2x - 4y + 5$.
[Figure: A circle with radius 2 centered at $(1, 2)$, which is tangent to the x axis at $(1, 0)$.]
8. R is the unit disk; $\rho(x, y) = 3x^4 + 6x^2y^2 + 3y^4$.
[Figure: A circle with radius 1 and center the origin.]
9. R is the region enclosed by the ellipse $x^2 + 4y^2 = 1$; $\rho(x, y) = 1$.
[Figure: An ellipse with center the origin, major axis 2, and minor axis 1.]
10. $R = \{(x, y) | 9x^2 + y^2 \leq 1, x \geq 0, y \geq 0\}$; $\rho(x, y) = \sqrt{9x^2 + y^2}$.
[Figure: The quarter section of an ellipse in the first quadrant with center the origin, major axis 2, and minor axis roughly 0.64.]

11. R is the region bounded by $y = x, y = -x, y = x + 2, y = -x + 2$; $\rho(x, y) = 1$.
[Figure: A square with side length square root of 2 rotated 45 degrees, with corners at the origin, $(0, 2)$, $(1, 1)$, and $(\text{negative } 1, 1)$.]
12. R is the region bounded by $y = \frac{1}{x}, y = \frac{2}{x}, y = 1$, and $y = 2$; $\rho(x, y) = 4(x + y)$.
[Figure: A complex region between 2 and 1 that sweeps down and to the right with boundaries $y = 1/x$ and $y = 2/x$.]

In the following exercises, consider a lamina occupying the region R and having the density function ρ given in the preceding group of exercises. Use a computer algebra system (CAS) to answer the following questions.

- (a) Find the moments M_x and M_y about the x -axis and y -axis, respectively.
 - (b) Calculate and plot the center of mass of the lamina.
 - (c) [T] Use a CAS to locate the center of mass on the graph of R .
13. [T] R is the triangular region with vertices $(0, 0)$, $(0, 3)$, and $(6, 0)$; $\rho(x, y) = xy$.
 14. [T] R is the triangular region with vertices $(0, 0)$, $(1, 1)$, and $(0, 5)$; $\rho(x, y) = x + y$.
 15. [T] R is the rectangular region with vertices $(0, 0)$, $(0, 3)$, $(6, 3)$, and $(6, 0)$; $\rho(x, y) = \sqrt{xy}$.
 16. [T] R is the rectangular region with vertices $(0, 1)$, $(0, 3)$, $(3, 3)$, and $(3, 1)$; $\rho(x, y) = x^2y$.
 17. [T] R is the trapezoidal region determined by the lines $y = -\frac{1}{4}x + \frac{5}{2}, y = 0, y = 2$, and $x = 0$; $\rho(x, y) = 3xy$.
 18. [T] R is the trapezoidal region determined by the lines $y = 0, y = 1, y = x$, and $y = -x + 3$; $\rho(x, y) = 2x + y$.
 19. [T] R is the disk of radius 2 centered at $(1, 2)$; $\rho(x, y) = x^2 + y^2 - 2x - 4y + 5$.
 20. [T] R is the unit disk; $\rho(x, y) = 3x^4 + 6x^2y^2 + 3y^4$.
 21. [T] R is the region enclosed by the ellipse $x^2 + 4y^2 = 1$; $\rho(x, y) = 1$.
 22. [T] $R = \{(x, y) | 9x^2 + y^2 \leq 1, x \geq 0, y \geq 0\}$; $\rho(x, y) = \sqrt{9x^2 + y^2}$.
 23. [T] R is the region bounded by $y = x, y = -x, y = x + 2$, and $y = -x + 2$; $\rho(x, y) = 1$.
 24. [T] R is the region bounded by $y = \frac{1}{x}, y = \frac{2}{x}, y = 1$, and $y = 2$; $\rho(x, y) = 4(x + y)$.

In the following exercises, consider a lamina occupying the region R and having the density function ρ given in the first two groups of Exercises.

- (a) Find the moments of inertia I_x, I_y , and I_0 about the x -axis, y -axis, and origin, respectively.
 - (b) Find the radii of gyration with respect to the x -axis, y -axis, and origin, respectively.
25. R is the triangular region with vertices $(0, 0)$, $(0, 3)$, and $(6, 0)$; $\rho(x, y) = xy$.
 26. R is the triangular region with vertices $(0, 0)$, $(1, 1)$, and $(0, 5)$; $\rho(x, y) = x + y$.
 27. R is the rectangular region with vertices $(0, 0)$, $(0, 3)$, $(6, 3)$, and $(6, 0)$; $\rho(x, y) = \sqrt{xy}$.
 28. R is the rectangular region with vertices $(0, 1)$, $(0, 3)$, $(3, 3)$, and $(3, 1)$; $\rho(x, y) = x^2y$.

29. R is the trapezoidal region determined by the lines $y = -\frac{1}{4}x + \frac{5}{2}$, $y = 0$, $y = 2$, and $x = 0$; $\rho(x, y) = 3xy$.
30. R is the trapezoidal region determined by the lines $y = 0$, $y = 1$, $y = x$, and $y = -x + 3$; $\rho(x, y) = 2x + y$.
31. R is the disk of radius 2 centered at $(1, 2)$; $\rho(x, y) = x^2 + y^2 - 2x - 4y + 5$.
32. R is the unit disk; $\rho(x, y) = 3x^4 + 6x^2y^2 + 3y^4$.
33. R is the region enclosed by the ellipse $x^2 + 4y^2 = 1$; $\rho(x, y) = 1$.
34. $R = \{(x, y) | 9x^2 + y^2 \leq 1, x \geq 0, y \geq 0\}$; $\rho(x, y) = \sqrt{9x^2 + y^2}$.
35. R is the region bounded by $y = x$, $y = -x$, $y = x + 2$, and $y = -x + 2$; $\rho(x, y) = 1$.
36. R is the region bounded by $y = \frac{1}{x}$, $y = \frac{2}{x}$, $y = 1$, and $y = 2$; $\rho(x, y) = 4(x + y)$.
37. Let Q be the solid unit cube. Find the mass of the solid if its density ρ is equal to the square of the distance of an arbitrary point of Q to the xy -plane.
38. Let Q be the solid unit hemisphere. Find the mass of the solid if its density ρ is equal to the distance of an arbitrary point of Q to the origin.
39. The solid Q of constant density 1 is situated inside the sphere $x^2 + y^2 + z^2 = 16$ and outside the sphere $x^2 + y^2 + z^2 = 1$. Show that the center of mass of the solid is not located within the solid.
40. Find the mass of the solid $Q = \{(x, y, z) | 1 \leq x^2 + z^2 \leq 25, y \leq 1 - x^2 - z^2\}$ whose density is $\rho(x, y, z) = k$, where $k > 0$.
41. [T] The solid $Q = \{(x, y, z) | x^2 + y^2 \leq 9, 0 \leq z \leq 1, x \geq 0, y \geq 0\}$ has density equal to the distance to the xy -plane. Use a CAS to answer the following questions.
 - (a) Find the mass of Q .
 - (b) Find the moments M_{xy} , M_{xz} , and M_{yz} about the xy -plane, xz -plane, and yz -plane, respectively.
 - (c) Find the center of mass of Q .
 - (d) Graph Q and locate its center of mass.
42. Consider the solid $Q = \{(x, y, z) | 0 \leq x \leq 1, 0 \leq y \leq 2, 0 \leq z \leq 3\}$ with the density function $\rho(x, y, z) = x + y + 1$.
 - (a) Find the mass of Q .
 - (b) Find the moments M_{xy} , M_{xz} , and M_{yz} about the xy -plane, xz -plane, and yz -plane, respectively.
 - (c) Find the center of mass of Q .
43. [T] The solid Q has the mass given by the triple integral $\int_{-1}^1 \int_0^{\frac{\pi}{4}} \int_0^1 r^2 dr d\theta dz$. Use a CAS to answer the following questions.
 - (a) Show that the center of mass of Q is located in the xy -plane.
 - (b) Graph Q and locate its center of mass.

44. The solid Q is bounded by the planes $x + 4y + z = 8$, $x = 0$, $y = 0$, and $z = 0$. Its density at any point is equal to the distance to the xz -plane. Find the moment of inertia I_y of the solid about the xz -plane.
45. The solid Q is bounded by the planes $x + y + z = 3$, $x = 0$, $y = 0$, and $z = 0$. Its density is $\rho(x, y, z) = x + ay$, where $a > 0$. Show that the center of mass of the solid is located in the plane $z = \frac{3}{5}$ for any value of a .
46. Let Q be the solid situated outside the sphere $x^2 + y^2 + z^2 = z$ and inside the upper hemisphere $x^2 + y^2 + z^2 = R^2$, where $R > 1$. If the density of the solid is $\rho(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$, find R such that the mass of the solid is $\frac{7\pi}{2}$.
47. The mass of a solid Q is given by $\int_0^{2\sqrt{2}} \int_0^{\sqrt{8-x^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{16-x^2-y^2}} (x^2 + y^2 + z^2)^n dz dy dx$, where n is an integer. Determine n such that the mass of the solid is $(2 - \sqrt{2})\pi$.
48. Let Q be the solid above the cone $x^2 + y^2 = z^2$ and below the sphere $x^2 + y^2 + z^2 - 4kz = 0$. Its density is a constant $k > 0$. Find k such that the center of mass of the solid is situated 7 units from the origin.
49. The solid $Q = \{(x, y, z) | 0 \leq x^2 + y^2 \leq 16, x \geq 0, y \geq 0, 0 \leq z \leq x\}$ has the density $\rho(x, y, z) = k$. Show that the moment M_{xy} about the xy -plane is half of the moment M_{yz} about the yz -plane.
50. The solid Q is bounded by the cylinder $x^2 + y^2 = a^2$, the paraboloid $b^2 - z = x^2 + y^2$, and the xy -plane, where $0 < a < b$. Find the mass of the solid if its density is given by $\rho(x, y, z) = \sqrt{x^2 + y^2}$.
51. Let Q be a solid of constant density k , where $k > 0$, that is located in the first octant, inside the circular cone $x^2 + y^2 = 9(z - 1)^2$, and above the plane $z = 0$. Show that the moment M_{xy} about the xy -plane is the same as the moment M_{yz} about the yz -plane.
52. The solid Q has the mass given by the triple integral $\int_0^1 \int_0^{\pi/2} \int_0^{r^2} (r^4 + r) dz d\theta dr$.
- Find the density of the solid in rectangular coordinates.
 - Find the moment M_{xy} about the xy -plane.
53. The solid Q has the moment of inertia I_x about the yz -plane given by the triple integral $\int_0^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} \int_{\frac{1}{2}(x^2+y^2)}^{\sqrt{x^2+y^2}} (y^2 + z^2)(x^2 + y^2) dz dx dy$.
- Find the density of Q .
 - Find the moment of inertia I_z about the xy -plane.
54. The solid Q has the mass given by the triple integral $\int_0^{\pi/4} \int_0^{2\sec\theta} \int_0^1 (r^3 \cos\theta \sin\theta + 2r) dz dr d\theta$.
- Find the density of the solid in rectangular coordinates.
 - Find the moment M_{xz} about the xz -plane.
55. Let Q be the solid bounded by the xy -plane, the cylinder $x^2 + y^2 = a^2$, and the plane $z = 1$, where $a > 1$ is a real number. Find the moment M_{xy} of the solid about the xy -plane if its density given in cylindrical coordinates is $\rho(r, \theta, z) = \frac{d^2 f}{dr^2}(r)$, where f is a differentiable function with the first and second derivatives continuous and differentiable on $(0, a)$.

56. A solid Q has a volume given by $\iint_D \int_a^b dz dA$, where D is the projection of the solid onto the xy -plane and $a < b$ are real numbers, and its density does not depend on the variable z . Show that its center of mass lies in the plane $z = \frac{a+b}{2}$.
57. Consider the solid enclosed by the cylinder $x^2 + z^2 = a^2$ and the planes $y = b$ and $y = c$, where $a > 0$ and $b < c$ are real numbers. The density of Q is given by $\rho(x, y, z) = f'(y)$, where f is a differential function whose derivative is continuous on (b, c) . Show that if $f(b) = f(c)$, then the moment of inertia about the xz -plane of Q is null.
58. [T] The average density of a solid Q is defined as $\rho_{ave} = \frac{1}{V(Q)} \iiint_Q \rho(x, y, z) dV = \frac{m}{V(Q)}$, where $V(Q)$ and m are the volume and the mass of Q , respectively. If the density of the unit ball centered at the origin is $\rho(x, y, z) = e^{-x^2-y^2-z^2}$, use a CAS to find its average density. Round your answer to three decimal places.
59. Show that the moments of inertia I_x, I_y , and I_z about the yz -plane, xz -plane, and xy -plane, respectively, of the unit ball centered at the origin whose density is $\rho(x, y, z) = e^{-x^2-y^2-z^2}$ are the same. Round your answer to two decimal places.